Week No. 2

Lab Manual: Information Security

Lab Title: Classical Ciphers (Code)

Lab Objectives

By the end of this lab, students will be able to:

4. Perform cryptanalysis on classical substitution ciphers.
5. Use frequency analysis to identify cipher patterns.
6. Demonstrate the process of recovering plaintext without knowing the key (Brute force).

Learning Outcomes (CLOs & PLOs Mapping)

CLO	CLO Statement	Bloom's Level	PLO
1	Apply Information Security techniques into practice.	P3 (Apply)	PLO-5

Lab Overview

This lab introduces Cryptanalysis, the art and science of breaking ciphers without knowing the key.

Students will practically explore frequency analysis, a statistical technique used to infer key or plaintext patterns from ciphertexts of monoalphabetic ciphers such as Caesar.

Experiment 1: Understanding Frequency Distribution

Theory:

Every language has a characteristic letter frequency.

For example, in English, the letters E, T, and A appear most frequently.

Letter Frequency (%)

E 12.7

T 9.1

A	8.2
O	7.5
I	7.0
N	6.7

Task:

1. Take a sample text (plaintext).
2. Count the occurrence of each alphabet letter.
3. Plot or print the frequency table.

Pseudo Code:

```
INPUT: Plaintext
INITIALIZE dictionary for A-Z counts
FOR each character in Plaintext:
   IF character is alphabet:
      Convert to uppercase
      Increment count in dictionary
CALCULATE percentage frequency of each letter
DISPLAY frequency table
```

- **Expected Output:**
  A table showing each letter's frequency percentage.

Experiment 2: Frequency Analysis on Caesar Cipher

Theory:
Since Caesar Cipher is a simple shift cipher, the **frequency distribution** of the ciphertext matches that of plaintext — only shifted.

By identifying the most frequent letter in ciphertext and comparing it to expected frequencies (like 'E'), the **shift key** can be deduced.

Pseudo Code:

```
INPUT: Ciphertext
CALCULATE frequency of each letter
```

```
IDENTIFY most frequent letter (assume it represents 'E')
ESTIMATE key = (cipher_letter - 'E') mod 26
DECRYPT using estimated key
OUTPUT possible plaintext
```

Task:

- Ciphertext: "SOJZKXS KDGS OY ZUU KGYE"
- Perform frequency analysis to find the most probable key and recover plaintext.

Experiment 3: Frequency Analysis on Substitution Cipher

Theory:
A simple substitution cipher maps each plaintext letter to a unique ciphertext letter.

Frequency analysis helps guess the mapping by comparing observed frequencies to English frequency charts.

Pseudo Code:

```
INPUT: Ciphertext
CALCULATE frequency of letters in ciphertext
SORT letters by frequency (descending)
COMPARE with English frequency order (ETAOINSHRDLU...)
CREATE possible substitution mapping
APPLY mapping to ciphertext
OUTPUT partial or complete plaintext
```

Task:

- Use the following ciphertext and attempt partial decryption:

  ZOLSS AOL JHU KPFF TVYF AOHA AOPZ PZ H ZLTWSL JYPWALY

(Hint: Try matching most frequent ciphertext letter to ‘E’.)

Experiment 4: Automated Frequency Counter

Objective:

Write a small Python/C++ function to automatically count letter frequency and suggest possible Caesar shifts.

Pseudo Code:

```
for key in range(26):
    decrypted = ""
    for char in ciphertext:
        if char.isalpha():
            decrypted += chr(((ord(char) - 65 - key) % 26) + 65)
        else:
            decrypted += char
    print("Key:", key, "=>", decrypted)
```

Task:

- Use brute-force + frequency intuition to find the correct decryption.